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RiceKernelEngine: Benchmarking transfer learning models for microscopic images of rice kernel

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ABSTRACT

Rice serves as a principal dietary staple food nationwide. The high demand for this cereal grain has led to extensive research, resulting in the frequent development of new grain varieties. Proper cultivation time, region, and nurturing techniques specific to each rice variety can significantly boost production. However, the subtle differences distinguishing one rice grain from another make the classification of rice kernels a resource-intensive and exhaustive task. Our proposed approach explores the potential for automated classification of five different rice varieties developed in Bangladesh: BINADHAN-8, BINADHAN-23, BRRI-67, BRRI-74, and BRRI-102. These varieties are highly similar in appearance, and state-of-the-art transfer learning models have been employed to assess their practical feasibility. The dataset comprises 3155 images, with classes containing 605, 578, 642, 695, and 635 images, respectively. The implemented models—VGG19, MobileNetV3, EfficientNet, and ConvNeXt—achieved accuracies of 85%, 94%, 96%, and 87%, respectively, with an average accuracy of 90.50%. The results indicate the practical applicability of these models in this field, with EfficientNet providing the highest accuracy rate for classifying rice grains. This study utilized an exclusive, self-obtained dataset of microscopic rice grain images. An autonomous and efficient classification system will benefit rice farmers and open new research opportunities for agriculturists and other stakeholders.

1. Introduction

Rice is a highly produced and consumed grain worldwide. It is the core ingredient in the vast majority of Asian cuisines as well as international dishes. With its neutral flavor and versatility, rice pairs easily with a wide variety of dishes, from meats to vegetables. As a result, from 2008–2009 to 2021–2022, the quantity of rice consumed increased significantly, from 437.18 million to 520 million metric tons, showing an upward trend of 82.82 million metric tons as depicted in Fig. 1, according to a survey conducted by Statista [1]. China and India hold the top positions as rice-producing countries, with China cultivating over 148 million metric tons and India producing 129 million metric tons in the 2021–2022 year [2]. In 2022–2023, China achieved a record-breaking rice production of 155 million metric tons, while India, as the second highest-ranked country, produced 108.5 million metric tons [3]. A worldwide overview of rice consumption and production for the year 2022–2023, according to Statista, is presented in Fig. 1.

Bangladesh is the third highest rice-consuming country worldwide, according to a survey conducted by Statista [3]. Rice is a staple in Bangladesh, with each citizen consuming 144.5 kg per year [4], and 67.5% of the daily caloric intake comes from rice [5]. In 2021, Bangladesh produced 56.9 million metric tons of rice paddy. Since the country's independence in 1972, annual rice production has increased by 2.89% as of 2021 [6]. According to a report by the Daily Sun, Bangladesh was ranked as the third-largest rice-producing country globally for the fourth consecutive time by the Food and Agriculture Organization (FAO) in 2022 [7]. Rice contributes 70% of Bangladesh's agricultural GDP, and one-sixth of the national income is derived from it [8].

This significant demand for grain refers to continuous research resulting in the invention of a massive number of rice varieties. The reason is that the purity of a rice variety is a price factor while trading, making classification of the varieties an important task. But these varieties may seem almost indistinguishable to the untrained human eye. Consequently, making the task of classification not only

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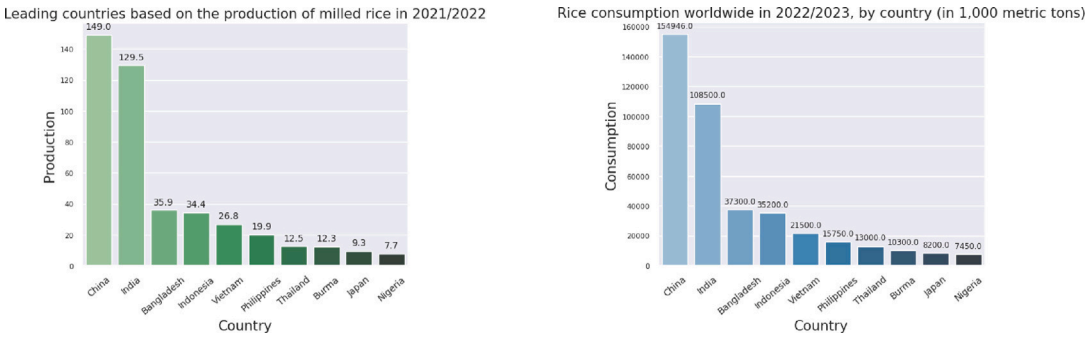


Fig. 1. (A) Leading countries for milled rice production, in 2021/2022 (B) Rice Consumption Worldwide, in 2022/2023.

time and resource-heavy task, but also leaves massive room for human error. Thus, making manual classification unfeasible and inconsistent. Nonetheless, an autonomous classification system can overcome the aroused limitations.

Although the use of machine learning is still not widely practiced in the agricultural sector, automating the classification of rice varieties could significantly address challenges in this field. An automated process using image processing and machine learning methods would be faster, more cost-effective, and more consistent. Additionally, there are numerous deep-learning models available that can address this issue to varying degrees. In this study, we incorporated several popular transfer learning models and conducted a detailed comparison to identify and propose the method best suited for efficiently classifying rice varieties.

The key contributions of this study are:

1. Developing an autonomous system to classify rice varieties quickly and efficiently.
2. Achieving a higher accuracy rate in classification with a unique dataset.

Following the introduction, related studies and research connected with our study are briefly discussed in the Related Works section. After that, data collection, data pre-processing, and model training are presented in the Methodology section. The Results and Discussion section evaluates and compares the accuracy rates and performance of the models. Finally, we conclude the paper in the Conclusions section.

2. Related works

With the vision of making rice grain classification an automated system, numerous studies have been proposed over the years. In 2005, Zhao-yan and Fang [9] developed an approach to classify six varieties of rice found in China using neural networks. Principal Component Analysis (PCA) was implemented to reduce feature dimensions, and a two-layer tan-sigmoid method was applied for classification. Their implemented technique achieved an accuracy ranging from 74% to 95%.

In 2010, researchers collected another dataset of Chinese rice, consisting of five different rice seed varieties from Jiangxi Agricultural University. A backpropagation classification technique was applied to classify the rice grain species, achieving an accuracy of 93.66% [10]. Despite the achieved accuracy, the datasets in [9,10] lack robustness.

Kuo and Chung [11] conducted research to classify a total of 30 types of rice from 16 different countries, including Iran, Pakistan, Taiwan, India, Bangladesh, and others. Fifty grains from each variety of rice seed were sampled, and a sparse representation-based classification technique was implemented. This technique achieved an average accuracy of 89.1%. Despite the dataset's versatility, high-resolution images of rice seeds (0.95 pixels per mm) are required to attain this accuracy level.

Similarly, a classification method for 52 rice varieties from five agro-ecological zones of the Philippines was implemented by Guzman and Peralta [12]. Thirteen features were extracted from the rice varieties, and a four-layer Multi-Layer Perceptron was employed, achieving an overall accuracy of 70%. Nevertheless, higher accuracy could potentially be achieved by using varied image sizes and additional pre-processing techniques.

Subsequently, in 2013, [13,14], and [15] proposed three different methods for classifying rice varieties. In [13], 250 grain samples from nine different rice types were used. A Multi-Layer Perceptron and Combined Feature Model were implemented to classify the varieties, with the Combined Feature Model achieving a high accuracy of 92%. Similarly, Gujjar and Siddappa [14] examined six basmati rice varieties and applied a Backpropagation Neural Network, achieving accuracies of 90%, 88%, 95%, 82%, 74%, and 80% for each type, respectively. In contrast, Kaur and Singh [15] implemented a Multi-Class Support Vector Machine (SVM) for rice classification, achieving an accuracy of 86%. However, this approach is not time-efficient for classifying rice kernels.

In another study [4], four types of rice—White Basmati, Red Basmati, White Samba, and Red Samba—were classified using the RGB color model. This model successfully classified both white and red rice grains with nearly perfect accuracy. In May 2014, rice variety detection using Principal Component Analysis was proposed in the study [16]. This study achieved an overall accuracy of 79% in classifying three types of Basmati rice. However, the achieved accuracy is lower than that of [14]. Another study on Basmati rice grain classification employed image processing and a neural network pattern recognition tool, achieving a 98.7% accuracy rate by removing certain morphological features in the pre-processing stage [17].

In the same year, Yammen and Rityen [18] conducted a study to classify Thai white rice using image processing techniques. To detect grain boundaries, the Canny operator was used, and the Euclidean method determined the length of the rice grain. This study achieved a 99.33% accuracy in classification, considering grain length only. In 2020, another method was proposed to classify five different Thai rice grains—RD6, RD15, RD23, RD75, and M105. A mean average precision of 0.8 was achieved using a region proposal-based convolutional neural network [19].

Research has also been conducted in India on the classification of different rice varieties. In 2017, a group of researchers proposed a unique approach for classifying 13 distinct milled rice categories found in Tamil Nadu, India [20]. Methods such as Multi-Layer Perceptron (MLP), Support Vector Machine (SVM), Decision Tree (DT), and Bayesian Network were applied to five shape and size features, 48 color features, and eight textural properties derived from the dataset. The highest accuracy, 92.30%, was achieved with the Multi-Layer Perceptron. Another study conducted by Bhensjaliya and Vasava [21] used a Pattern Recognition Neural Network with eight-layer input neurons to classify 268 rice grains of six varieties: Basmati, Chhapi, Gujarat −17, Jirasar, Masoori, Parimal, and Ponia. With this method,

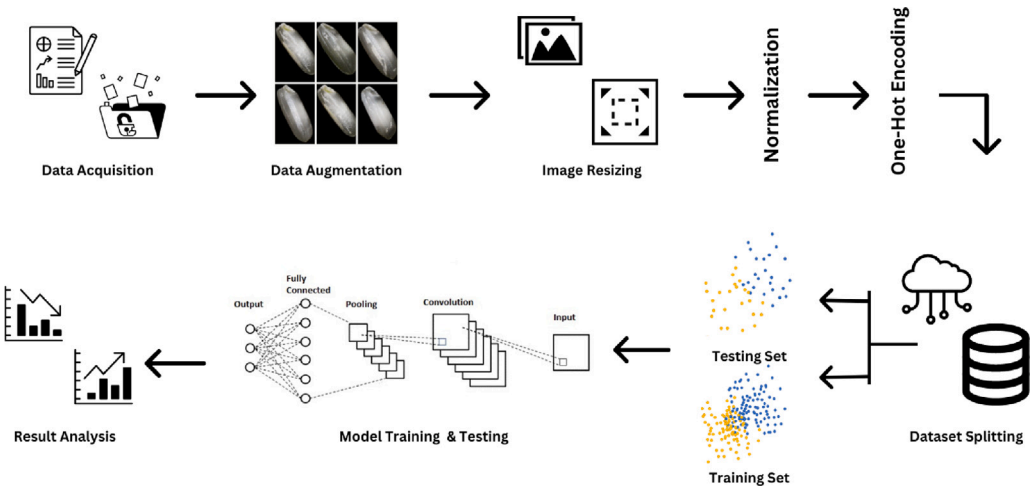


Fig. 2. Workflow Diagram.

an average accuracy of 75% was obtained. However, the accuracy could be improved by using different sampling and sample pre-processing techniques.

Additionally, a significant number of research studies have been conducted at various times on different types of rice grains produced in Turkey. In November 2015, a method was proposed to classify three Turkish rice grain categories—Baldo, Osmancik, Yasemin—and broken rice using machine learning and image processing techniques [22]. The Nearest Neighbor algorithm achieved an accuracy of 91.03%, which was the highest among the approaches tested. To achieve this accuracy level, rice images needed to be taken under well-lit laboratory conditions, avoiding natural light.

In 2019, a study applied various machine learning algorithms to 3810 images of two rice species found in Turkey to classify them [23]. Among all the models tested, the Linear Regression model achieved the highest accuracy at 93.02%. Despite this high accuracy level, the dataset includes only two species of Turkish rice varieties.

From 2021 to 2022, studies [24,25], and [26] were conducted on a dataset consisting of five different rice varieties from Turkey: Arborio, Ipsala, Basmati, Karacadag, and Jasmine. These studies were based on 15,000 images from each rice variety, from which 13 morphological features, 90 color features, and 4 shape features were extracted. The proposed method in [26] involved the application of machine learning techniques, including Linear Regression, Multi-Layer Perceptron, Support Vector Machine, Decision Tree, Random Forest, Naïve Bayes, and K-Nearest Neighbour, to classify the rice varieties.

Incorporating new features resulted in a higher accuracy rate for the models. Among all the techniques, the Multi-Layer Perceptron model performed the best, achieving an accuracy of 99.91%. On the same dataset, Prakash and Rajkumar [24] applied a Vanilla Neural Network and VGG16, achieving accuracies of 95.39% and 99.58%, respectively. However, in 2021, Cinar and Koklu [25] achieved the highest classification accuracy on this dataset using deep learning methods. The obtained accuracies were 99.87%, 99.95%, and 100% for the Artificial Neural Network (ANN), Binarized Neural Network (BNN), and Convolutional Neural Network (CNN), respectively, although the dataset includes only five rice varieties produced in Turkey.

Moreover, research on this subject has also been conducted in other countries, such as Malaysia and Myanmar. In particular, Pazoki and Farokhi [27] applied two artificial neural network methods—Multi-Layer Perceptron and Fuzzy Method—on five Iranian rice varieties. The overall classification accuracy achieved in this study was 99.46%.

Additionally, in a study by Tin and Mon [28], image processing techniques were used to classify Myanmar rice grains. Five morphological features were extracted from 38 rice grains of five Myanmar rice varieties, achieving an accuracy of 80% in classification. Higher accuracy could be achieved by incorporating other methods along with feature analysis for classification. Table 1 outlines the research discussed above.

The compared reviews of relevant studies on rice classification, there are few major challenges that has not been combated by existing studies. A few of them are:

- Dataset limitation by both region and quantity
- Scarcity of research on microscopic rice image, and single grain per image
- Essentially, lack of research for Bangladeshi-produced and developed rice varieties

Principally, this paper focuses on the five categories of rice grains that are developed in Bangladesh by "Bangladesh Institute of Nuclear Agriculture (BINA)" and "Bangladesh Rice Research Institute (BRRI)" named- BINADHAN-8, BINADHAN-23, BRRI-67, BRRI-74, BRRI-102. The uniqueness and self-obtained dataset makes this paper stand out from others.

3. Methodology

This particular section offers an in-depth overview of the methods of data collection followed by the general chronology of a machine learning-based study: data collection, data pre-processing, and training models. The workflow diagram of the whole process has been condensed in Fig. 2.

3.1. Data collection

For this study, a comparatively different dataset was needed to provide the uniqueness of it. In order to fulfill the objective, 3155 microscopic colored images of five different classes named as: BINADHAN-8, BINADHAN-23, BRRI-67, BRRI-74, and BRRI-102 were taken. The rice grains were gathered firsthand from the Bangladesh Institute of Nuclear Agriculture (BINA) and the Bangladesh Institute of Rice Research Institute (BRRI) with their husks intact. The grains were then manually hulled, without any machinery tool assistance to extract the rice kernels. Manual hulling was necessary to avoid damaging the embryo and keeping it as close to its natural state as possible.

To collect the images, a digital microscope with image resolution up to 1920X1080, a focus range of 15 mm–40 mm, and 1600X magnification was used to take the snapshots of each rice grain. Firstly, any

Table 1
Rice grain classification studies.

Title	Dataset	Models	Result	Limitations
[16]	3 types of Basmati Rice - Classic, Rozana, Rice	Principal Component Analysis	Overall accuracy rate: 79%	Low accuracy rate, small sample size
[24]	5 Turkish rice varieties - Arborio, Ipsala, Karacadag, Jasmine, Basmati (Total 75000 images)	Vanilla Neural Network, VGG16	Vanilla Neural Network Accuracy: 95.39%, VGG16 Accuracy: 99.58%	Dataset only includes 5 Turkish rice varieties
[13]	250 grain sample of 9 Sri Lankan rice varieties	Multi-Layer Perceptron, Combined Feature Model	Multi-Layer Perceptron Accuracy: 34%, Combined Feature Model Accuracy: 92%	Different sampling methods could yield higher accuracy
[29]	4 types of rice: White Basmati, Red Basmati, White Samba, Red Samba	RGB Color Model	Red Grains Accuracy: 100%, White Grains Accuracy: 100%	Only works for a small sample of colored rice
[22]	3 types of Turkish rice grains - Baldo, Osmancik, Yasemin Broken rice	12 Machine Learning Algorithms: NNge, NB Tree, RBF Networks, K Star, BF Tree, Bagging, Random Forest, J48, IB1, IBk, JRip, Naïve Bayes	Classification Accuracy: 89.46% to 91.03% across various models	Not efficient for natural or low-light rice images
[20]	13 types of milled rice from Tamil Nadu, India	MLP, SVM, DT, BN	MLP Accuracy: 92.20%, SVM Accuracy: 90.384%, DT Accuracy: 59.615%, BN Accuracy: 82.692%	Dataset includes only milled rice varieties
[12]	52 rice varieties from 5 Agro-ecological Zones of the Philippines	4-layer MLP	Accuracy: 70%	Dataset includes only Philippine varieties; different sampling methods and larger samples could improve accuracy
[28]	5 types of Myanmar rice grains	Image Processing Techniques	Overall Accuracy: 80%	Classification accuracy could be optimized
[18]	Binary rice images	Canny Operator for grain boundary detection, Euclidean method for grain length measurement	Average Accuracy: 99.33%	Considers rice length only for classification
[14]	6 varieties of Basmati rice	Backpropagation Neural Network	Classification Accuracy: 74% to 95% across varieties	Dataset includes Basmati rice varieties only
[30]	90 images of 3 rice types	Multi-Class SVM	Classification Accuracy: 92.22%	Feature and classification techniques need precision for accuracy
[15]	10 geometric features of rice	Multi-class SVM	Classification Accuracy: 86%	Classification process is time-consuming
[23]	3810 images of 2 rice species	LR, MLP, SVM, DT, RF, NB, KNN	Highest Accuracy: 93.02% for LR	Larger dataset and more features could improve accuracy
[9]	6 Chinese rice varieties	PCA, Two-layer tan-sigmoid method	Classification Accuracy: 74% to 95% across varieties	High similarities among grains decrease accuracy
[21]	2688 rice grains of 6 types	Pattern Recognition Neural Network	Overall Accuracy: 75%	Different sampling methods could boost accuracy
[11]	30 types of rice from 16 countries	Sparse-Representation-Based Classification	Average Accuracy: 89.1%	High-resolution grain images are required
[10]	5 Chinese rice types	Backpropagation Classification	Classification Accuracy: 82.82% to 99.99% across varieties	Dataset lacks robustness
[17]	Basmati rice grains	Neural Network, Pattern Recognition	Overall Accuracy: 98.7%	Some features were removed in the pretreatment process
[27]	5 Iranian rice varieties	ANN, Neuro-Fuzzy	Overall Accuracy: 99.4%	Limited number of rice varieties

damaged or diseased rice was separated from strong and sturdy grains. After that, 110–150 rice grains of each type of rice were taken and each rice was photographed in 6 different alignments using the timed shot setting. Approximately, 50–100 images from each class were taken focusing on the embryo to make the classification process easier. As shown in Fig. 3, rice grains were laid on a velvet sheet cloth to get the

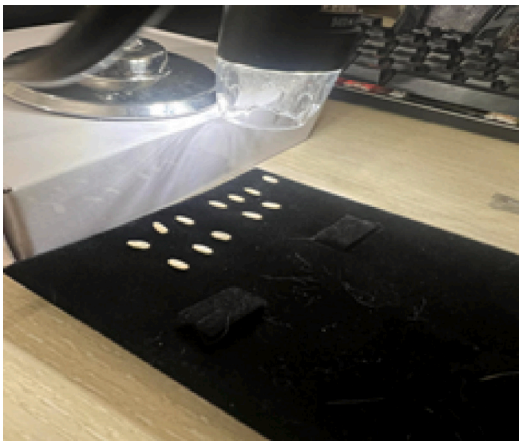


Fig. 3. Data collection process.



Fig. 4. From left to right (a)–(e): Sample Images from Dataset. (a) BRRI-67 (b) BRRI-102 (c) BINADHAN-8 (d) BINADHAN-23.

Table 2
Images collected for each class.

Class name	Number of images
BINADHAN-8	605
BINADHAN-23	578
BRRI-67	642
BRRI-74	695
BRRI-102	635

perfect jet-black background in order to emphasize more on the grain textures.

A total of 3,155 images were collected, with representative samples from each class illustrated in Fig. 4. The class-wise distribution of images is summarized in Table 2, comprising BINADHAN-8 (605 images), BINADHAN-23 (578 images), BRRI-67 (642 images), BRRI-74 (695 images), and BRRI-102 (635 images). The number of images per class ranges from 578 (lowest) to 695 (highest). Given the minimal variation in class sizes, the dataset exhibits only a slight imbalance, which was deemed negligible; therefore, no additional techniques were applied to address it.

For pre-processing, We then split the total dataset for training and testing sets, with a ratio of 80% (2524 samples) and the rest 20% (631 samples) accordingly. No separate validation data were kept. Testing data was used for both validation and testing.

Table 3
Example image with class label.

Image Vector	Class Label
[0, 0, 0, 1, 0]	BRRI-74

3.2. Data pre-processing

Fig. 4 illustrate data pre-processing procedures step by step. Since the classes of data are balanced and there is no garbage data, no intensive preprocessing steps were needed. All steps taken are standard procedures in image classification tasks. The steps are image augmentation, resizing, normalization, and label one-hot encoding, not strictly in those order.

3.2.1. Augmentation

Data augmentation refers to the training dataset size increment process with artificial methods to add a new version from the existing data. Besides increasing dataset size, it also increases variations and randomness in data that would be present in unseen real-world data. Thus making the data more generalized and decreasing the chance of overfitting. In this work, every image was vertically and horizontally flipped and rotated in a range of 90 degrees.

3.2.2. Resizing

Images are resized to 224 × 244 pixels. It is necessary since the augmented images can be of diverse shapes, however, the deep learning models demand a uniform shape for all images. This image size is determined by the image quality with existing process resources. A model's higher resolution image training would be more hardware-intensive and time-consuming compared to lower ones. Resizing the images to a lower resolution does result in some loss of information however it does not leave a huge impact on training.

3.2.3. Normalization

Image normalization is also a common pre-processing step for computer vision and image processing. Used for standardizing an image's pixel values with scaling to a specific range, for easy employing and ineffective lighting condition variation quality of the model. The equation is depicted as shown.

$$P(x,y) = MIN + \frac{(P_{xy} - min_{x,y}) \cdot (MAX - MIN)}{max_{x,y} - min_{x,y}} \quad (1)$$

Here, $P(x,y)$ is the normalized pixel value at (x,y) coordinate.

P_{xy} known as pixel value comprising (x,y) coordinate before normalization.

$min_{x,y}$ indicates that the image's pixel value is minimum.

$max_{x,y}$ refers to the image's pixel value being maximum. MAX and MIN values are the new range in which the normalized values should be. set to 1 and 0 respectively, in this case.

3.2.4. Label encoding

Finally, labels undergo a process called one-hot encoding where this technique involves the conversion of categorical to binary vectors, with each vector's length equaling a complete number of categories. In this representation, only one vector element is set to 1 (indicating 'hot'), while all others are set to 0 (indicating 'cold'). For example (see Tables 3 and 4),

It is also a frequently used step for classification tasks. This concludes the preprocessing. Next, we move on to training.

Table 4
One-hot encoded vector representation.

Image	BINADHAN-8	BINADHAN-23	BRRI-67	BRRI-74	BRRI-102
[0, 0, 0, 1, 0]	0	0	0	1	0

3.3. Training models

For this classification, we used CNN (Convolutional Neural Networks). CNNs have multiple layers, each trained to distinguish different features of the input images. Usually, the layers start off with simple features, and as the layers progress, the filters become increasingly complex. Each layer requires a unique identification of a set of features in the images, which outputs convolved images that then become the input for the next layer. CNNs can be used on prebuilt models and can be retrained with new datasets, giving CNNs an advantage for image classification tasks. Typically, a CNN has 3 layers: a Convolutional layer, a Pooling layer, and a Fully Connected layer.

The two matrices—one consisting of learnable parameters (the kernel) and the other highlighting the constrained receptive field segments—execute a dot product in the convolutional layer. Consider having a neuron layer with a size of $N \times N$ square, followed by our convolutional layer. When employing an $m \times m$ filter ω, our convolutional layer's output will be equivalent to the size of $(N-m+1)\times(N-m+1)$. For the pre-nonlinearity calculation, we take input to some unit x_{ij}^{ℓ} to aggregate the advancements (weighted by the filter elements) in the preceding layer cells.

The convolutional layer operation is defined as follows:

$$x_{ij}^{l} = \sum_{a=0}^{m-1}\sum_{b=0}^{m-1} \omega_{ab} y_{(i+a)(j+b)}^{(l-1)} \tag{2}$$

The convolutional layer then applies its nonlinear function:

$$y_{ij}^{l} = \sigma(x_{ij}^{l}) \tag{3}$$

The pooling layer essentially decreases the spatial size of the convolved feature and extracts dominant features. The FC or Fully connected layer has full connectivity with preceding and succeeding layers. It can be evaluated by a matrix multiplication after a bias effect, mapping input to output layers. The pre-trained CNN models which we incorporated in our proposed classification strategy, are VGG19, MobileNetV3, ConvNeXt, and EfficientNet.

For further tuning the models and stopping the model from overfitting or underfitting, Adam optimizer was used. It combines the idea of Root Mean Square Propagation (RMSP) and the momentum of Gradient Descent, first proposed at a conference in 2015 [31].

The equation below showcases the updated rule for the Adam optimizer.

$$m_t = \beta_1 \cdot m_{t-1} + (1-\beta_1)\cdot g_t \tag{4}$$

$$v_t = \beta_2 \cdot v_{t-1} + (1-\beta_1)\cdot g_t^2 \tag{5}$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t} \tag{6}$$

$$\hat{v}_t = \frac{v_t}{1-\beta_2^t} \tag{7}$$

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\hat{v}_t}+\epsilon}\cdot \hat{m}_t \tag{8}$$

Here, θ_t represents the parameters at iteration t. g_t is the gradient of the objective function with respect to the parameters at iteration t. α is the learning rate. β_1 and β_2 are the exponential decay rates for the first and second moments, respectively. m_t is the first moment (mean) estimate at iteration t. v_t is the second moment (uncentered variance) estimate at iteration t. $\hat{m}_t$ and $\hat{v}_t$ are bias-corrected estimates of the first as well as the second moments. ϵ, a small constant, (*e.g.*, 1×10^{-8}) which is summed to the denominator to ensure the arithmetic stability. values used for the hyperparameters can be found in Table 5.

Table 5
Hyperparameters table.

Optimizer	Adam
Learning Rate (α)	0.005
β_1 (β_1)	0.9
β_2 (β_2)	0.999
Epsilon (ϵ)	0.1
Decay 1	0.0
Epochs	50

Table 6
Parameter values for the `ReduceLROnPlateau` callback function.

Metric	val_accuracy
Patience	5
Factor	0.7
Verbose	1
Minimum Learning Rate	1e-4

A callback function named `ReduceLROnPlateau`, was used to decrease the learning rate by a certain factor if the monitored metric ceased to show improvement for a certain number of epochs.

- Monitor a specified metric (*e.g.*, `val_accuracy`) in the ongoing training procedure.
- If the monitored metric does not delineate any improvements after the number of epochs passes specified by `patience`, reduce the learning rate by multiplying it with the specified factor.
- A minimum value for the learning rate was defined to prevent the learning rate from becoming too small over time.
- The `verbose` parameter is used to control the amount of output during training.

The values used for the parameters for the callback function can be found in Table 6.

3.3.1. VGG19

The convolutional neural network consists of 19 layers, of which 16 and 3 are fully linked to categorize images into 1000 object categories. It is mainly used in the field of image and vision research. VGG19 is based on the ImageNet database containing 1 million images of him from 1,000 categories (Fig. 6). It adopts multiple 3 × 3 filters in each convolutional layer which are fully connected, aids in image identification [32]. With deeper layers into the network, the filter quantity grows, beginning with 64 filters in the first layer and then doubling the filters after the MaxPooling layer. After two convolutional layers max pooling performs 2 × 2 filters. After pulling layers the output passed through three fully connected layers. During training which is shown in Algorithm 1 and Fig. 5, the model learns how to optimize weights and biases with loss function minimization like the cross-entropy, which compares the probabilities of predicted class to ground truth labels. This optimization is performed using backpropagation and gradient descent algorithms [33].

Algorithm 1 Transfer Learning (VGG19) based Model

```
Begin
Load the VGG19 model and freeze the top layers
Execute the top layers
Pass the output to a GlobalAveragePooling2D layer
for i = 1 to 6 do
    Pass the output of the previous layer into a dense layer
end for
Pass the output of the previous layer using a Softmax activation
End
```

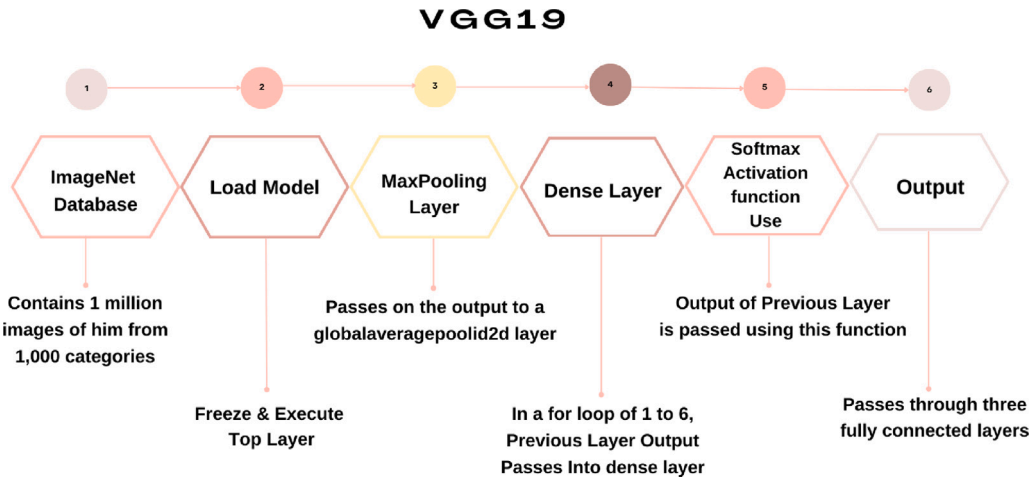


Fig. 5. Structure of Employed Model VGG19 [34].

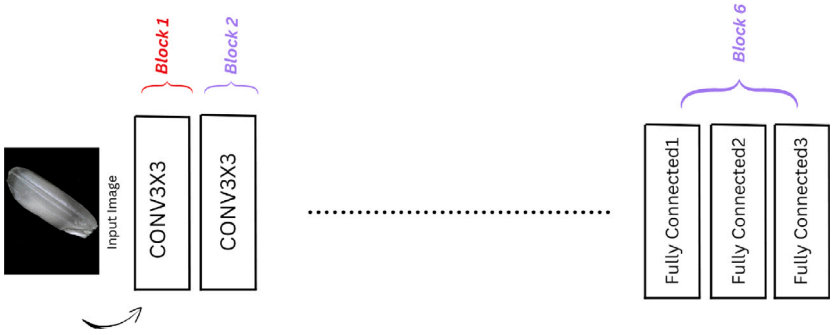


Fig. 6. Layers of VGG19.

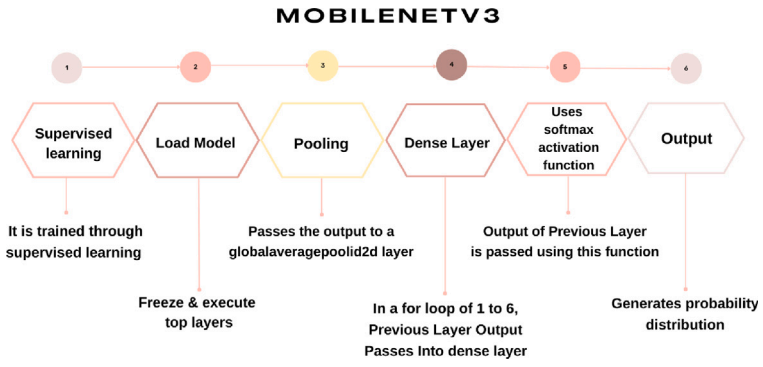


Fig. 7. Structure of Employed model MobileNetV3 [35].

3.3.2. *MobileNetv3*

This widely applied deep neural network model derives image characteristics across numerous convolutional layers (See Fig. 7). Here, the model passes an image as input through an initial convolutional layer with a tiny picture size to reduce the spatial dimensionality of the image. It consists of multiple blocks, each containing a combination of depth-separable turns and linear bottleneck layers. An inverse residual is a residual link with a bottleneck layer where the number of channels increases and then decreases again.

After the MobileNetV3 block, the feature maps are globally averaged to reduce the spatial dimension to 1×1. A completely linked plane is added over the global average pool attribute to perform the final classification shown in Algorithm 2. MobileNetV3 is trained through supervised learning. The model uses techniques such as depth-separable convolution, inverted residuals, and SE blocks to achieve a balance between accuracy and efficiency. Neural networks are becoming more popular as the image data amount gets processed utilizing mobile apps increases. However, these networks require a huge amount of computation and advanced hardware support, making them difficult to adapt to mobile devices. MobileNetV3 can achieve an optimal trade-off in efficiency evaluation with mobile devices executing classification tasks [36].

3.3.3. *ConvNeXt*

Notably, the depthwise convolution approach with a smaller parameter set performed more effectively than the Swin Transformer (Fig. 8).

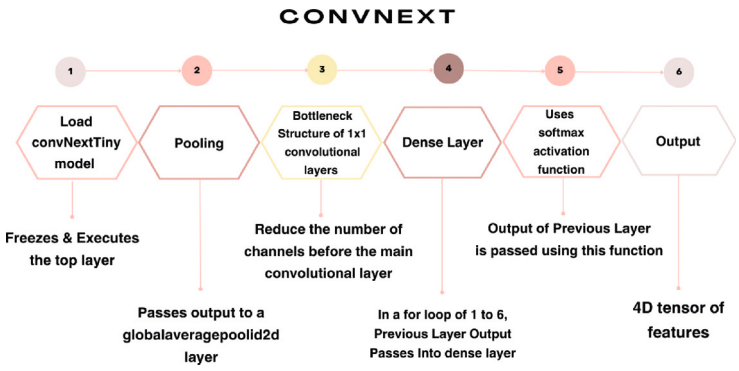


Fig. 8. Structure of Employed model ConvNeXtTiny [37].

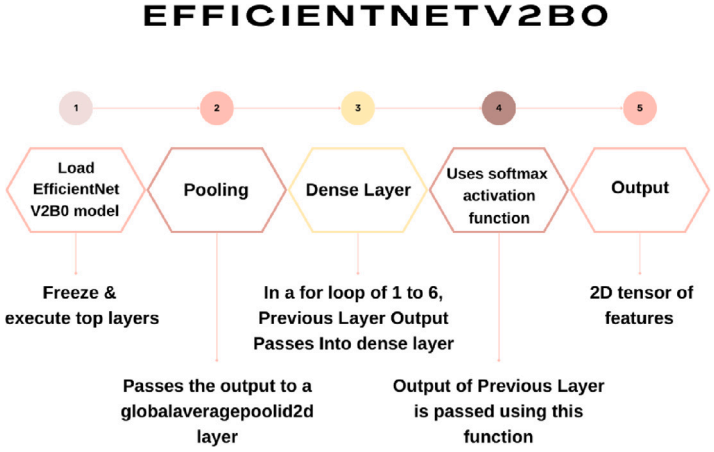


Fig. 9. Structure of Employed model EfficientnetV2b0 [38].

Algorithm 2 Transfer Learning (MobileNetV3Large) based Model

1: **Begin**
2: Load the MobileNetV3Large model and freeze the top layers
3: Execute the top layers
4: Pass the output to a GlobalAveragePooling2D layer
5: **for** $i = 1$ **to** 6 **do**
6: Pass the output of the previous layer into a dense layer
7: **end for**
8: Pass the output of the previous layer using a Softmax activation function
9: **End**

Algorithm 3 Transfer Learning (convNeXtTiny) based Model

1: **Begin**
2: Load the convNeXtTiny model and freeze the top layers
3: Execute the top layers
4: Pass the output to a GlobalAveragePooling2D layer
5: **for** $i = 1$ **to** 6 **do**
6: Pass the output of the previous layer into a dense layer
7: **end for**
8: Pass the output of the previous layer using a Softmax activation
9: **End**

Therefore, our study objectives are focused on the newly introduced ConvNeXt mesh, outperforming the Swin Transformer, and finding the best waste classification algorithm 3. ConvNeXt exceeds Swin Transformer regarding accuracy, performance, and simplicity [39]. This model adopts an inception-like structure with parallel branches and enables various network scaling features captured. The convNeXt uses a bottleneck structure consisting of 1×1 convolutional layers that reduce the channel number before the main convolutional layer. This model is trained through supervised learning. During training, the network learns about weights and biases optimization with a cross-entropy loss function minimization, with differentiating predicted class possibilities and ground truth labels. ConvNeXt is based on the ResNet structure [39].

3.3.4. *EfficientNetV2b0*

With an aim of advancing our work, we scaled a neural architecture base network, that initiated towards a model named EfficientNetV2b0, which provides significant accuracy and efficiency than earlier ConvNets. This model deals with depth, width, and resolution scaling. Combining all three aspects provides a better balance between model complexity and performance (Fig. 9). As presented in Algorithm 4, it starts with a basic network consisting of a set of convolutional layers, after a batch normalization function and a nonlinear activation function. In order to mitigate the computational complexity issue, EfficientNestV2b0 additionally performs depthwise separable convolutions. Basically, this model uses a compound scaling method that scales the depth, width, and resolution of the network. EfficientNetV2b0 is trained using supervised learning with backpropagation and gradient descent. Convolutional layers use different filter sizes, including 3×3, 5×5, and 7×7 filters. Overall, EfficientNetV2b0 is a powerful architecture for image classification tasks offering precise efficacy.

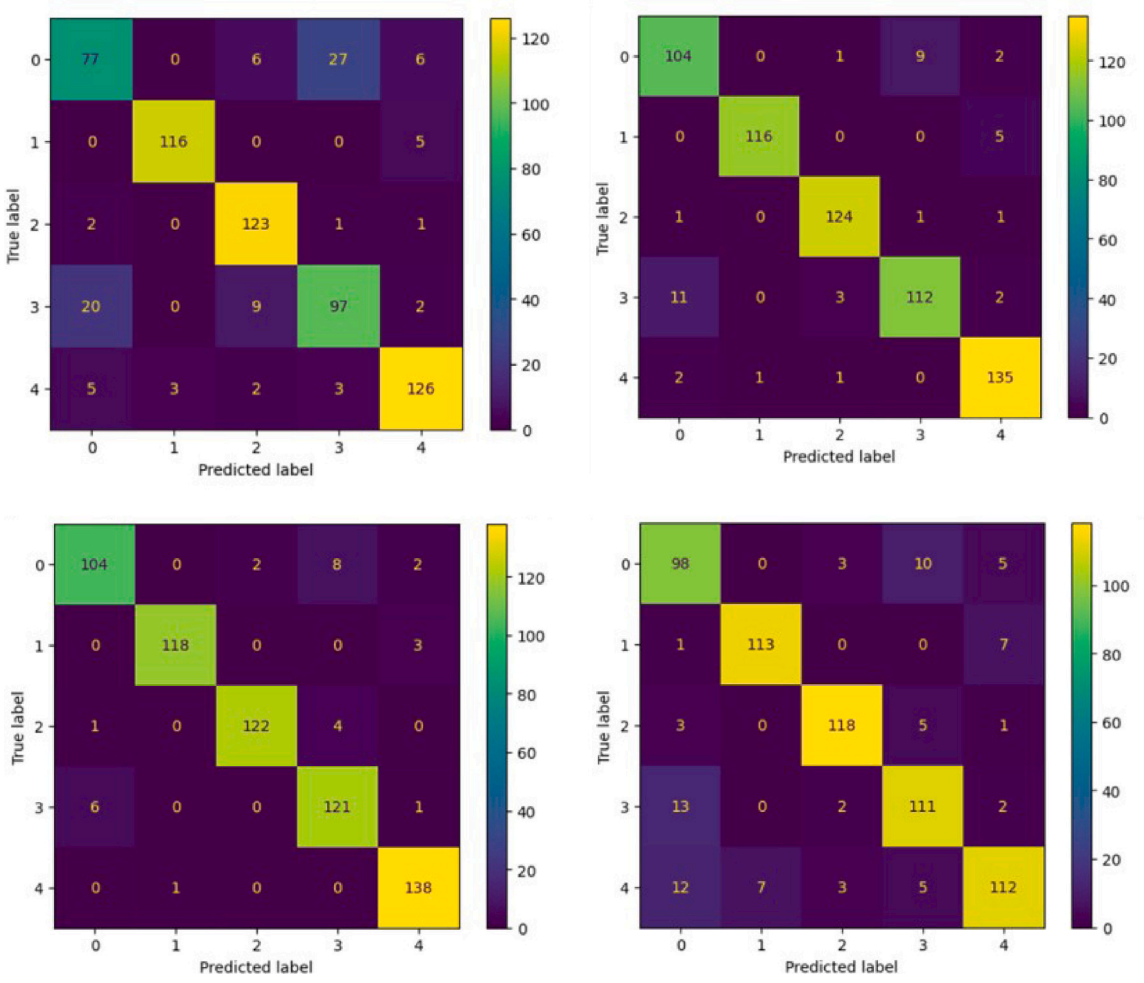


Fig. 10. From left to right (a)–(d): Confusion matrix of different models. (a) VGG19 (b) MobileNetV3Small (c) EfficientNetV2B0 and (d) ConvNeXtTiny.

Table 7
Accuracy of the transfer learning models.

Model	VGG19	MobileNetV3Small	EfficientNetV2B0	ConvNeXtTiny
Accuracy	85%	94%	96%	87%

The computer vision community presents it as a popular widespread choice [40].

Algorithm 4 Transfer Learning (EfficientNetV2B0) based Model

1: **Begin**
2: Load the EfficientNetV2B0 model and freeze the top layers
3: Execute the top layers
4: Pass the output to a GlobalAveragePooling2D layer
5: **for** $i = 1$ **to** 6 **do**
6: Pass the output of the previous layer into a dense layer
7: **end for**
8: Pass the output of the previous layer using a Softmax activation
9: **End**

4. Result and analysis

The comparative analysis conducted in this study involved the utilization of four transfer learning models: VGG19, MobileNetV3Small, EfficientNetV2B0, and ConvNeXtTiny. These models were evaluated based on their test accuracy metrics, which are presented in Table 7. The results demonstrate the exceptional performance of all four transfer learning models, with each achieving a test accuracy exceeding 85% (see Fig. 8).

The images used for training and evaluation were resized to a dimension of 224 × 224 pixels. The dataset consisted of five classes, namely BRRI-67, BRRI-102, BRRI-74, BINADHAN-23, and BINADHAN-8. The total number of data samples available for the study was 3155, with a subset of 631 samples designated for testing purposes. To ensure representative evaluation, the test data was stratified to maintain a proportional distribution of samples from each class.

In addition to accuracy, the evaluation of transfer learning models can be enriched by considering additional performance metrics such as precision, recall, and F1-score. We calculated precision with the division of total true and false positives with the number of true positives, revealing insights of the effectiveness of a model for precise positive instance identification. Conversely, the division of true positives by actual positive values is known as recall, reflecting the efficacy of the model in capturing all positive instances.

Mathematically, precision and recall can be represented as follows:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \tag{9}$$

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \tag{10}$$

To evaluate the F1-score, we utilize both precision and recall. F1-score's equation is presented below:

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

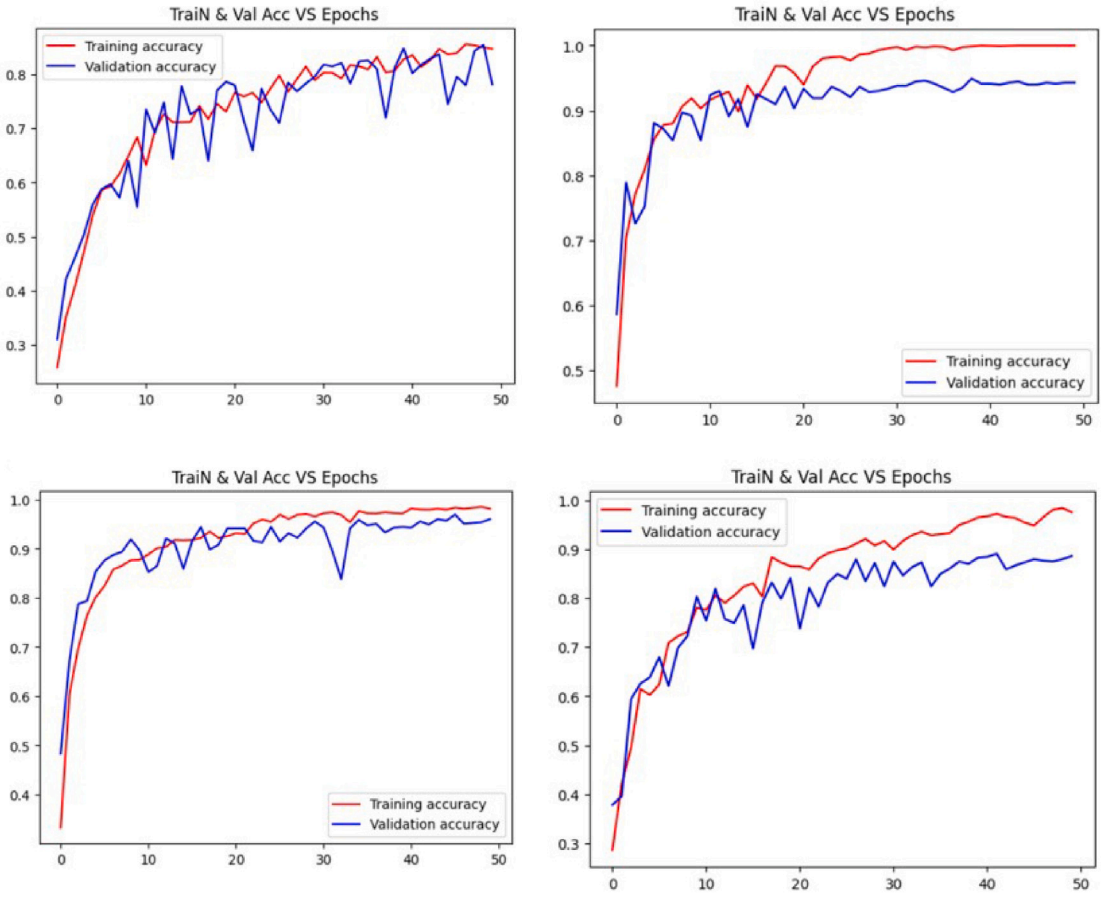


Fig. 11. From left to right (a)–(d): Plot between train and validation accuracy VS epochs. (a) VGG19 (b) MobileNetV3Small (c) EfficientNetV2B0 and (d) ConvNeXtTiny.

Table 8
Precision of the transfer learning models.

Models	VGG19	MobileNetV3Small	EfficientNetV2B0	ConvNeXtTiny
BRRI-67	0.76	0.92	0.91	0.85
BRRI-74	0.90	0.93	0.96	0.88
BRRI-102	0.88	0.96	0.98	0.94
BINADHAN-8	0.97	0.99	0.99	0.94
BINADHAN-23	0.74	0.88	0.94	0.77

Table 9
Recall of the transfer learning models.

Models	VGG19	MobileNetV3Small	EfficientNetV2B0	ConvNeXtTiny
BRRI-67	0.76	0.88	0.95	0.87
BRRI-74	0.91	0.97	0.99	0.81
BRRI-102	0.97	0.98	0.96	0.93
BINADHAN-8	0.96	0.96	0.98	0.93
BINADHAN-23	0.66	0.90	0.90	0.84

Table 10
F1-score of the transfer learning models.

Models	VGG19	MobileNetV3Small	EfficientNetV2B0	ConvNeXtTiny
BRRI-67	0.76	0.90	0.93	0.86
BRRI-74	0.90	0.95	0.98	0.84
BRRI-102	0.92	0.97	0.97	0.93
BINADHAN-8	0.97	0.97	0.98	0.94
BINADHAN-23	0.70	0.89	0.92	0.81

Integrating precision and recall [41] into the performance evaluation of transfer learning models enables a more comprehensive assessment of their classification abilities. While accuracy alone provides a measure of overall correctness, precision, and recall offers further model's potential assessment to correctly classify positive instances as well as retrieve relevant samples from the dataset. By considering these metrics collectively, a more nuanced understanding of the model's performance can be obtained, facilitating informed decision-making in various practical applications.

Table 8 presents the precision values for all models. Precision was calculated individually for all five labels in our classification. For BRRI-74, BRRI-102, and BINADHAN-23, EfficientNetV2B0 shows a much higher precision compared to the other models. All models showing great precision for BINADHAN-8 with MobileNetV3Small and EfficientNetV2B0 tying for the highest position with a score of 0.99. for BRRI-67, MobileNetV3Small shows the highest precision but not by a huge margin.

Table 9 presents the recall values for all models evaluated. Recall was calculated separately for each of the five class labels in the classification task. EfficientNetV2B0 achieved the highest recall for BINADHAN-8, BRRI-67, and BRRI-74. For BINADHAN-23, both MobileNetV3Small and EfficientNetV2B0 demonstrated the highest recall performance. For BRRI-102, MobileNetV3Small is the one with the highest recall score.

Table 10 presents the F1-score values for all models across the five rice varieties. F1-scores were computed individually for each class based on their corresponding precision and recall. Among the four models, EfficientNetV2B0 consistently achieved the highest or near-highest F1-scores across most classes. Notably, it shares the top performance on BRRI-102 with MobileNetV3Small, both achieving an F1-score of 0.97.

The confusion matrix (an N x N matrix) measures the classification capability of a model. Here, N denotes the total target class numbers. It

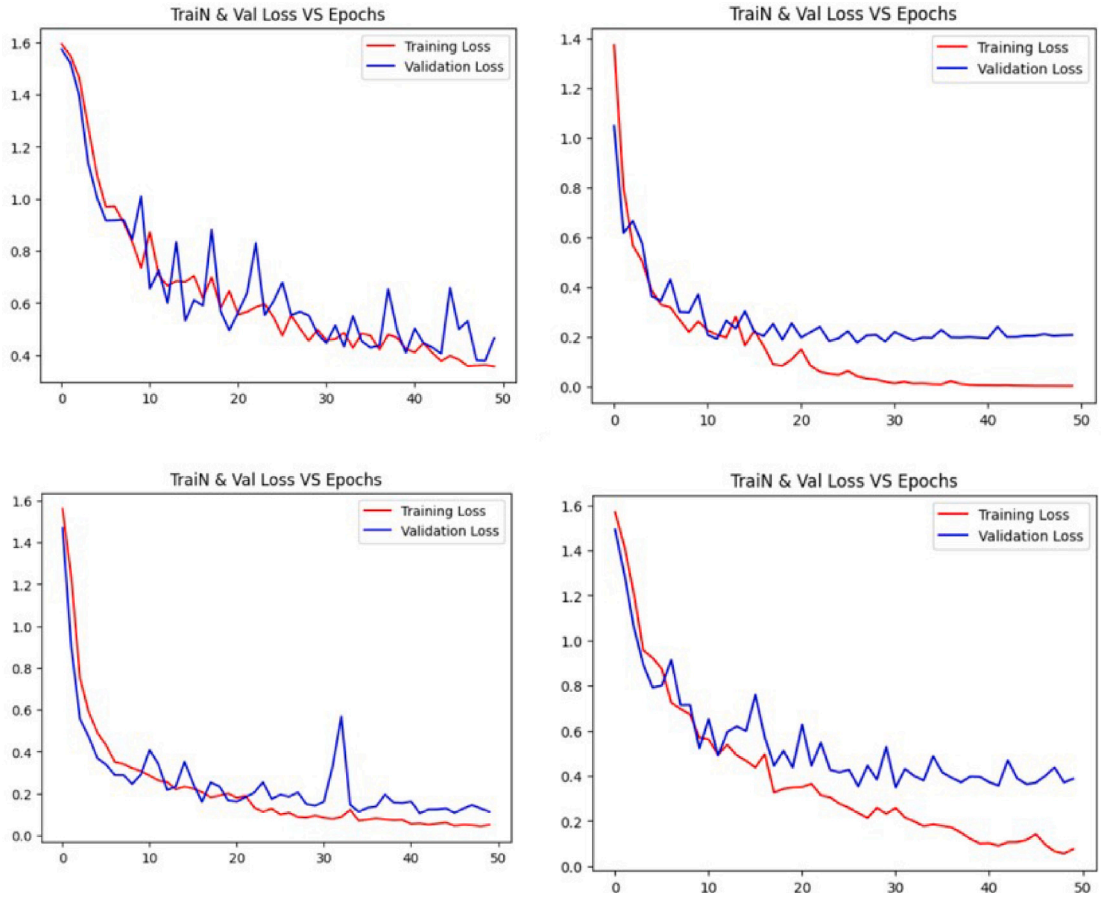


Fig. 12. From left to right (a)–(d): Plot between train and validation loss VS epochs. (a) VGG19 (b) MobileNetV3Small (c) EfficientNetV2B0 and (d) ConvNeXtTiny.

compares between prediction made by the model and the actual target values, to showcase a precise model's performance accuracy estimation and encountered error types [31].

Fig. 10 presents confusion matrices of each model. Here 0, 1, 2, 3, and 4 refer to the classes BINADHAN-23, BINADHAN-8, BRRI-102, BRRI-67, and BRRI-74 respectively. The majority of the models are struggling with class 0. EfficientNetV2B0, shown in Fig. 10(c), gives the highest result in class 1, 3, and 4 among all other models as shown in our previous calculations in precision. MobileNetV3Small shows the highest accuracy in class 2 but not by a big margin compared to VGG19 or EfficientNetV2B0. EfficientNetV2B0 also has the highest accurate classification in class 3 despite having lower precision than MobileNetV3Small.

To ensure an adequate learning process and convergence, the models underwent training for a total of 100 epochs. This duration allowed them to progressively refine their internal representations and optimize their performance.

Fig. 11 presents a graphical representation of the training and validation accuracy plotted against the number of epochs. For the VGG-19 model in Fig. 11(a), the validation accuracy falls within the range of 85%–90%. Similarly, the MobileNetV3Small model in Fig. 11(b) exhibits varying validation accuracy, spanning from 90% to 95%. Likewise, the EfficientNetV2B0 model in Fig. 11(c) also displays validation accuracy in the range of 90%–95%. Finally, the ConvNeXtTiny model in Fig. 11(d) demonstrates validation accuracy fluctuating between 85% and 90%. The consistent trend observed in the training and validation accuracies is noteworthy as it suggests that the models are performing according to expectations when presented with unseen data. This alignment between the two accuracy measures indicates that the models have effectively learned underlying patterns and can maintain similar levels of accuracy when faced with new data.

Here, Fig. 12 presents a graphical representation of the training and validation loss plotted with each epoch progression. Loss refers to penalization of inaccurate prediction, revealing the model's bad productivity on individual examples. Loss value zero indicates perfect performance while loss greater showcases faults in output. For precise accuracy, we need to evaluate the perfect weights and bias values that will generate the most minimal loss over all examples [42].

5. Real time application

Incorporating a real-time precise web application, we classified the five rice varieties. It was deployed in an AMD Ryzen 5 5600G processor including 6 cores at 3.9 GHz, with dual-channeled 8 GB RAM sticks and GPU(NVIDIA GTX 1660 Ti); functioning in Windows 11 Pro. Fig. 13 showcases a front-end comprising a simple user interface, we employed Next.js with a detailed image-based rice category display on the first page. Then a dynamic image upload page and assessment result as for the last page. Utilizing the FastAPI framework and Uvicorn for back-end API, the images get uploaded, after that, it runs the model for classification name.

6. Conclusion

To conclude, this research carried out state-of-the-art transfer learning models' in-depth assessment in classifying microscopic images of rice kernels. The significance of accurate and efficient classification in the agricultural industry cannot be overstated, as it directly impacts food quality, yield, and overall sustainability. It must also be highlighted that not only does this paper work with microscopic images of rice kernels which has never been done before but also the varieties shown in this paper are also unique varieties local to Bangladesh and

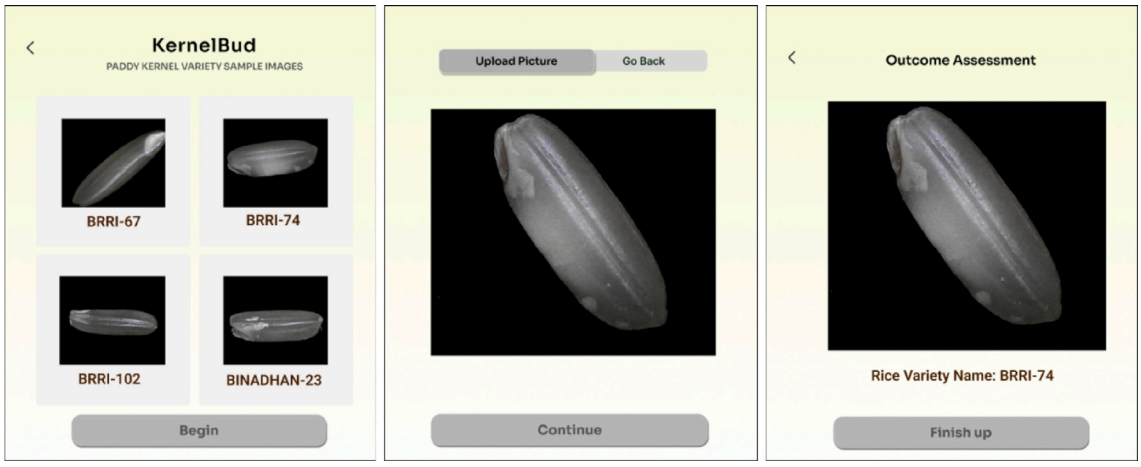


Fig. 13. Real-time web app for rice variety classification.

do not have any existing dataset. Leveraging transfer learning harnesses insights from pre-trained models proving an effective strategical way of tackling limited labeled data challenges and computational resources in this domain. The models we experimented with and evaluated in this work are VGG19, MobileNetV3Small, EfficientNetV2B0, and ConvNeXt-Tiny showcasing accuracy of 85%, 94%, 96%, and 87% respectively. EfficientNetV2B0 seems to be the clear winner in accuracy, and this does not change in the other evaluation metrics such as precision, recall, and f1 score. The main issue faced when training the model was overfitting in MobileNetV3Small and ConvNeXtTiny. It was solved through an early-stopping function that halted the training if performance became stagnant. Such issues would not arise if more data was available. Despite these shortcomings, all of the transfer learning models performed above 85% with the highest being 96% which proves the strength of these models yet still leaves room for improvements. Furthermore, the best model can be applied in a real-time application.

CRediT authorship contribution statement

Mehrab Islam Arnab: Writing – original draft, Methodology, Data curation, Conceptualization. **Anika Tabassum Nafisa:** Validation, Methodology, Conceptualization. **Md Tahsin:** Writing – original draft, Formal analysis, Data curation. **Md Monjor Morshed:** Writing – original draft, Conceptualization. **Maksura Binte Rabbani Nuha:** Writing – original draft, Validation, Conceptualization. **Md Sawkat Ali:** Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Mahamudul Hasan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Maheen Islam:** Writing – review & editing, Writing – original draft. **Taskeed Jabid:** Software, Resources, Project administration. **Mohammad Rifat Ahmmad Rashid:** Writing – review & editing, Writing – original draft, Visualization. **Mohammad Manzurul Islam:** Project administration, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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